

ENGINEER ROOTED IN AUTOMATION, MACHINE SOFTWARE, AND INDUSTRIAL SYSTEMS WITH EXPERIENCE ACROSS MANUFACTURING, ROBOTICS AND EMBEDDED TECHNOLOGIES. I BRING SYSTEM-LEVEL THINKING, I VALUE CLARITY AND CONTINUOUS IMPROVEMENT. I AM SEEKING ROLES THAT OFFERS STRONGER OWNERSHIP AND OPPORTUNITY TO DESIGN, BUILD OR ARCHITECT IMPACTFUL ENGINEERING SOLUTIONS - WHETHER IN SOFTWARE, ROBOTICS OR SYSTEMS-ARCHITECTURE.

EDUCATION

MS, ELECTRICAL & COMPUTER ENGINEERING (3.41/4)	DEC-17
TANDON SCHOOL OF ENGINEERING, NEW YORK UNIVERSITY	
BE, INSTRUMENTATION & CONTROLS, (7.04/10)	JUN-14
SVIT, GUJARAT TECHNOLOGICAL UNIVERSITY	

SKILLS

COMPUTER PROGRAMMING:	C/C++/CS, JAVA (DESKTOP & WEB), GIT, PYTHON (AI-AUTOMATION TOOLS, LIMITED HANDS-ON CODING), WEB(HTML, CSS, JS), DESKTOP(JAVAFX, AWT)
AUTOMATION:	MICRO-CONTROLLERS & SINGLE BOARD COMPUTERS, PLCs(LADDER-LOGIC, FBD), HMI/SCADA
INDUSTRIES SERVED:	MACHINERIES, MANUFACTURING, CONTROL PANELS, SOLAR ENERGY, EDUCATION

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER III - AUTOMATION, UNIVERSAL INSTRUMENTS, NY Sep 22 - present

- Developing and maintaining **Fuzion** platform software on regular basis.
- Engineered **SBS (Side Barcode Scanning)** using Keyence SR-X300 with Live View, traceability, error recovery and multi-bank support for Connectors with barcodes on side of the body.
- Upgraded existing **AMS** process from single point to 13-locations for higher accuracy on wafer.
- Built a **wafer-to-board mapping** system that converts customer wafer map formats into Universal Fuzion board template files for direct use in production.
- Developed **Selective Placement** with parts buffering, DFF-based Patch & Interposer categories, and avoiding-priority rules for optimal patch-interposer matching.
- Implemented **AVP** to allow user-defined fiducial finds during ECS, NPI and Production flows.
- Integrated Cognex QuickBuild (VPP) with **MCOS↔QBIT** TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform.
- Built a CS **Light Intensity Controller** software with multi-channel control, network connectivity and controller discovery for illumination management.
- Implemented **nested-tray** support for Multi-Rot feeders.
- Added **isosceles triangle** fiducial support.
- Implemented configurable **park-position** on Fuzion platform.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree.

AUTOMATION ENGINEER , NOR-CAL CONTROLS, CA Jan - Jul 22

- Developed automation software for utility-scale solar power plants
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

SOFTWARE ENGINEER - MACHINE AUTOMATION, MINI-CIRCUITS, NY Feb 18 - Nov 21

- Developed and maintained software for pick-and-place rotary machines with computer vision inspection.
- Implemented Reverse motion on rotary machines with Pressing-arm to enable EVT testing.
- Implemented the new shopping cart system for the online store <https://www.minicircuits.com/>
- Developed enterprise application software for internal engineering teams.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript

PRIMA AUTOMATION, AHMEDABAD, INDIA Jun 14 - Jan 16

DESIGN ENGINEER May 15 - Jan 16

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

QUALITY CONTROL ENGINEER Jun 14 - May 15

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

ACADEMIC WORK EXPERIENCE

DEVELOPER INTERN, REVMAX FLEET OPTIMIZATION, NY Sep - Dec 17

- Developed firmware for the SidewalkIQ sensor using Doppler radar to monitor sidewalk activity.
- Deployed sensors at multiple NYC locations and validated field performance
- Conducted R&D to improve data consistency and optimize storage and transfer to backend servers.
- Tech stack : · Arduino-C ·

INSTRUCTOR (ITEST PROGRAM), ROBOTICS & MECHATRONICS DPT., NYU Sep - Dec 17

- Mentored to Robotics Club Programming Team for competitions (First Robotics & Vex Robotics).
- Taught concepts of Computer Programming to Robotics Club Team.

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
 - Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
 - Wireless Robot Controller - Arduino: <https://github.com/sap586/WirelessRobotDay5Activity>